BTS –VI –04.16 – 0307 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

A

B. Tech. Degree VI Semester Examination April 2016

ME 601 INSTRUMENTATION AND CONTROL SYSTEMS
(*2006 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Differentiate first and second order systems.
(b) List different static and dynamic characteristics.
(c) Explain the principle and working of load cell.
(d) Explain temperature compensation in thermocouple.
(e) Define transfer function of a system. What is its significance?
(f) Explain state space modelling of a system. What are its advantages?
(g) Explain the notion of stability. How can stability be determined using R.H. criteria.
(h) List the properties of Root loci technique.

PART B

(4×15 = 60)

II. (a) Explain different classifications of instrumentation with examples. (8)
(b) Explain static calibration and determination of bias systematic error. (7)

OR

III. Explain mercury in glass thermometer as a first order instrument. (15)

IV. (a) What are rosettes? (5)
(b) What is gauge factor? Explain the classifications of strain gauges. (10)

OR

V. Explain the working of RTD with a neat sketch. How is compensation done in RTD? (15)

VI. The open loop transfer function of a unity feedback system is (15)

$$G(s)=\frac{k}{s(s+1)(s+2)}.$$

Determine the value of K such that the steady state error is less than 0.1 for unit ramp input.

OR

VII. Model an armature controlled DC motor in transfer function and in its state space approach. (15)

VIII. Draw the root locus of $G(s)\,H(s)=\dfrac{k}{s(s^2+2s+2)}$. Comment on the stability of the system. (15)

OR

IX. (a) Explain different components of control system. (5)
(b) Define gain margin and phase margin. How can the nature of stability be obtained from frequency domain approach. (10)

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